Mini Project Phase I-A on

IOT BASED SMART HELMET

Submitted in partial fulfillment for the degree of Bachelor of Technology
In Computer Engineering

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CERTIFICATE

This is to certify that Tanuja Gangurde, Ridam Shitole,Komal Tiwari have completed the Mini Project Phase I-A on the topic "IOT BASED SMART HELMET" satisfactorily in partial fulfillment for the Bachelor Degree in Computer Engineering under the guidance of Prof. Sanjay R Ranveer during the year 2024-2025 as prescribed by S.N.D.T. Women's University, Mumbai.

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Chapter 1

Introduction

Motorcycle riders are highly vulnerable to accidents, with head injuries and delayed emergency response being the primary causes of fatalities. Despite the enforcement of traffic laws, challenges such as riding without helmets, drunk driving, and late medical assistance remain unresolved. To address these concerns, technology-driven solutions are essential. The integration of Internet of Things (IoT) enables smart helmets to go beyond traditional protection by detecting helmet usage, monitoring alcohol levels, identifying accidents, and sending instant alerts through GPS and GSM modules. This ensures not only prevention but also timely intervention during critical situations.

Alongside IoT, the use of Artificial Intelligence (AI) significantly enhances the effectiveness of smart helmets. By analyzing data from sensors such as accelerometers, vibration sensors, and breath analyzers, AI algorithms like Random Forest and Support Vector Machines (SVM) can predict accident scenarios with improved accuracy.

1.1 Motivation and Problem Statement

The alarming rise in two-wheeler accidents has become a pressing global issue. According to the World Health Organization (WHO), motorcyclists are among the most at-risk road users, with head injuries being the leading cause of death. Despite strict regulations mandating helmet usage, many riders neglect safety practices due to discomfort, negligence, or overconfidence. Moreover, even when helmets are worn, traditional models serve only as passive protective gear and lack the ability to actively prevent accidents or ensure timely medical assistance. This gap motivates the development of a Smart Helmet system that can actively contribute to saving lives.

One of the critical problems in road safety is drunk driving and helmet avoidance, both of which drastically increase accident risks. Conventional helmets cannot address these behaviors, but with the use of IoT and sensors, it becomes possible to monitor alcohol levels, verify helmet usage, and link these conditions directly with bike ignition systems. Furthermore, in the event of a crash, delays in providing medical aid significantly reduce survival rates. Current accident response systems are often dependent on human reporting, which may not always be reliable or immediate. This highlights the need for a system capable of automatic accident detection and emergency communication.

Beyond IoT-enabled monitoring, the inclusion of Artificial Intelligence (AI) offers a solution to improve decision-making and accident prediction. AI algorithms can process sensor data, recognize patterns of risky behavior, and predict accident scenarios with higher accuracy, thereby reducing false alarms and enhancing system reliability. This combination of IoT and AI creates a proactive safety framework that not only responds to accidents but also prevents them before they occur. The motivation behind this project is therefore rooted in leveraging modern technologies to address long-standing problems in road safety, ensuring both prevention and timely intervention for riders.

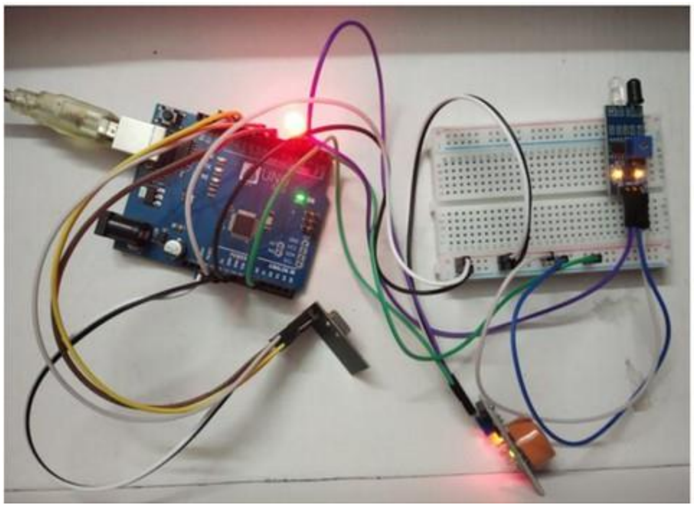
Chapter 2

Basic Block Diagram

2.1 Working

- System Initialization – On power-up, the microcontroller activates and calibrates all sensors, including helmet detection, alcohol sensing, accelerometer, GPS, and GSM modules.
- Helmet Verification – A limit switch or IR sensor ensures that the rider is wearing the helmet. If the helmet is not detected, the ignition system remains locked.
- Alcohol Screening – The MQ-3 sensor checks the rider's breath for alcohol concentration. If alcohol is above the permissible level, ignition is disabled and the rider is alerted.
- Ignition Authorization – When both safety checks are passed (helmet worn and rider sober), the helmet unit communicates with the bike unit, enabling the ignition relay.

- Continuous Monitoring – During the ride, the accelerometer, gyroscope, and vibration sensors track motion and detect abnormal shocks or crashes.
- Accident Detection – In case of an accident, the system confirms impact severity and immediately gathers GPS coordinates.
- Emergency Communication – The GSM module automatically sends an alert message with location details to predefined emergency contacts and, if required, to nearby hospitals or authorities.
- AI-Enhanced Decision Making – Artificial Intelligence algorithms such as Random Forest and Support Vector Machines (SVM) analyze multi-sensor data to distinguish genuine accidents from false triggers, ensuring accurate and timely response.



1. Microcontroller (Arduino)

The heart of the Smart Helmet system is the Arduino microcontroller (Arduino Nano / Mega 2560 / ESP32). It serves as the central processing unit that collects real-time data from various sensors, applies programmed logic, and manages communication with external modules. Arduino is chosen for its low cost, simplicity, large community support, and ability to handle multiple sensor inputs simultaneously. Its flexibility allows integration with IoT platforms and machine learning models for further enhancement.

2. Sensors

Helmet Detection Sensor (IR/Limit Switch)

Ensures that the helmet is worn before the motorcycle ignition is enabled. This enforces compliance with safety rules.

Alcohol Sensor (MQ-3 Gas Sensor)

Detects the concentration of alcohol in the rider's breath. If alcohol is above the permissible threshold, the ignition system is locked, preventing unsafe riding.

Accelerometer & Gyroscope (3-axis MEMS Sensor)

Monitors the rider's movement and orientation. Sudden shocks, falls, or abnormal tilts are interpreted as possible accidents.

Vibration/Force Sensor

Provides additional confirmation of a collision event by measuring external impacts and vibrations. GPS Module (Neo-6M)

Captures the real-time geographic coordinates of the rider. In case of an accident, these coordinates are shared with emergency contacts.

GSM Module (SIM800L)

Sends SMS alerts automatically to family members, hospitals, or authorities after detecting an accident.

3. Artificial Intelligence enhances the efficiency of the Smart Helmet by improving accident detection accuracy, with Random Forest and Support Vector Machine (SVM) being two widely used algorithms.

Random Forest, an ensemble method of decision trees, is highly effective in handling noisy and complex sensor data such as accelerometer spikes, vibration intensity, and gyroscope tilt. By aggregating the results of multiple trees, it minimizes false alarms and provides probability-based accident predictions, making it suitable for risk scoring. On the other hand, SVM works by identifying the optimal hyperplane that separates accident data from normal riding patterns. It is particularly useful when dealing with smaller, well-labeled datasets and ensures precise classification of events in real time. While Random Forest is more robust against outliers and large datasets, SVM is efficient for early deployment phases with limited training data. Together, these algorithms complement each other—SVM offering quick classification and Random Forest providing reliable, probability-driven predictions—making the Smart Helmet's AI system both accurate and adaptive.

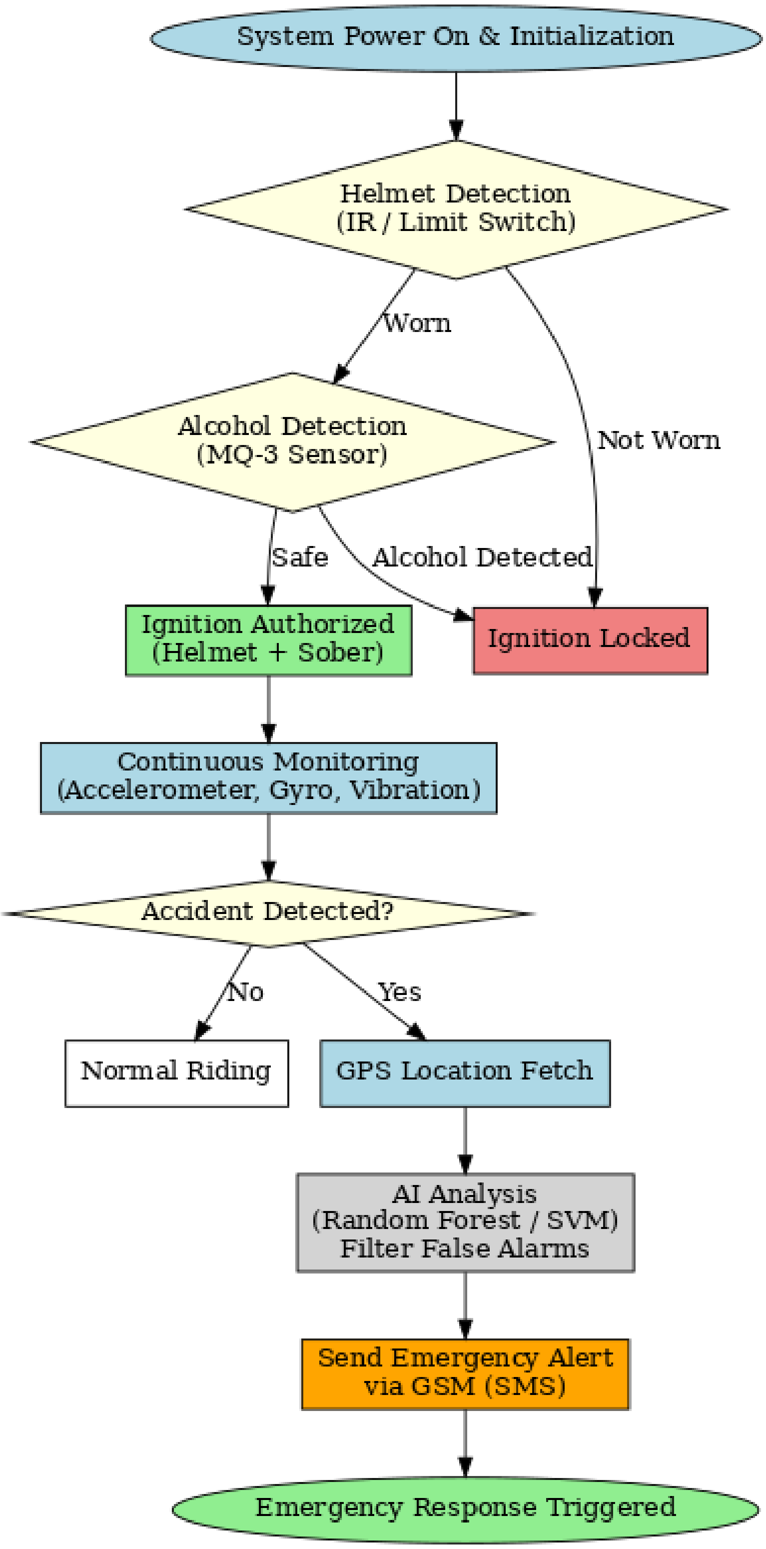


Figure 2.3: Flow Chart

Figure 2.3: System Architecture

Chapter 3 Literature Survey

3.1 Review of various papers

3.2

Paper Name	Author(s)	Year	Pros	Cons	Description
Design and Development of AI - IoT Based Smart Helmet to Ensure Safe Driving	Authors not clearly listed in snippet (IEEE Paper)	2023	Integration of AI with IoT; detects alcohol and helmet use; ensures safe driving.	Relies heavily on GSM/GPS; performance may degrade in low-signal areas.	This paper presents a smart helmet that integrates AI with IoT modules for rider safety. It ensures helmet usage, detects alcohol consumption, and uses GSM/GPS for accident alerts.
IoT Based Smart Helmet and Accident Identification System	Md. Atiqur Rahman, Toufiq Ahmed, S.M Ahsanuzzaman, Abid Ahsan, Ishman Rahman	2020	Accident identification with IR, alcohol detection, and accelerometer; integrates mobile application for alerts.	Depends on central database and network availability; limited scalability and data security concerns.	Proposes a helmet and bike unit connected via sensors and mobile application. Prevents drunk driving, ensures helmet usage, and provides accident location alerts.

Enhancing Human Safety Through IoT-Integrated Smart Helmet Technology	Venkatesh K, Yatheesh Kumar Reddy K, Harshitha Jodu, Sumalatha R, Sree Kanthi P	2024	Focuses on proactive safety; enforces helmet usage and alcohol detection; integrates GPS/GSM for emergency alerts.	Uses Arduino Uno with limited processing; solar dependency affects reliability.	This study emphasizes reducing accidents through IoT integration. It uses gas sensors for alcohol detection, limit switches for helmet verification, and vibration sensors for accident alerts.
Design & Implementation of IoT Based Smart Helmet for Road Accident Detection	Authors not clearly listed in snippet (IEEE Paper)	2022	Detailed design of IoT-enabled helmet; includes accident detection and GSM/GPS alert system.	Scalability challenges; limited to helmet-bike integration only.	Focuses on designing an IoT helmet system that enforces helmet use, detects accidents using sensors, and communicates location using GSM/GPS.
Analysis of Smart Helmets and Designing an IoT based Smart Helmet: A Cost Effective Solution	Authors not clearly listed in snippet (IEEE Paper)	2021	Cost-effective design; explores different helmet technologies and optimizations.	Primarily conceptual analysis; lacks large-scale implementation details.	Provides comparative analysis of existing smart helmet solutions and proposes a cost-effective IoT-based design focusing on affordability and safety.

3.3

Paper Name	Methodology	Key Features	Limitations
AI - IoT Based Smart Helmet (Safe Driving)	AI (Random Forest) + sensors (force, vibration, alcohol, accelerometer), GSM alerts	AI-based accident prediction; Alcohol detection; SOS via GSM; Real-time prediction (~70%)	Accuracy limited; GSM dependency; No direct emergency integration
IoT Based Smart Helmet & Accident Identification	Helmet (IR + alcohol); Bike (accelerometer, relay, Bluetooth); Mobile app + database	Helmet & alcohol detection; Accident detection via accelerometer; Mobile app alerts	Database issues; Needs smartphone app; IR sensor by passable
Enhancing Human Safety Through IoT-Integrated Helmet	Arduino + limit switch, alcohol & vibration sensors, GSM + GPS	Helmet & alcohol check before ignition; Vibration accident detection; GSM+GPS alerts	Low processing power (Arduino Uno); GSM/GPS dependency; No AI prediction
IoT Based Smart Helmet for Road Accident Detection	Arduino Nano + Mega; IR sensor; MQ-3 alcohol; vibration sensor; RF + GSM/GPS	Helmet detection via IR; Alcohol & accident detection; GPS SMS alerts; Compact circuits	Complex prototype; Comfort issues; Higher cost
Analysis & Design of IoT Smart Helmet (Cost Effective)	Survey + design proposal; alcohol, position, GPS, piezo crash sensor; IoT modem; solar panel	Cost-effective; Helmet + alcohol detection; Crash detection + GPS alerts; Solar power	Added weight; Solar insufficient; Concept only (no prototype)

Figure 3.1: Comparing Previous Papers

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Figure 3.1: Comparing Previous Papers

Chapter 4

Project Road Map

Figure 4.1: Project Roadmap

Chapter 5 Implementation

5.1 Methodology

Step 1: Requirement Analysis

The project begins with an in-depth study of road safety concerns, especially issues faced by two-wheeler riders such as not wearing helmets, riding under the influence of alcohol, and delays in emergency respons after accidents. These challenges were translated into specific requirements: (a) detection of helmet usage before engine ignition, (b) alcohol detection through breath sensors, (c) accident detection using motion an vibration sensors, (d) real-time location tracking using GPS, (e) emergency alerting through GSM, and (f) integration of Artificial Intelligence (AI) to improve accuracy and minimize false alarms. Hardware and software requirements were finalized, including the selection of Arduino/ESP32 microcontrollers, GSM/GP modules, MQ-3 gas sensor, accelerometer, gyroscope, and appropriate development platforms such as Arduino IDE and Python.

Step 2: System Design

The design phase involved creating a block diagram and a system architecture to map out how each hardware and software component interacts. The system was divided into two major units: the Helmet Unit, responsible for monitoring the rider's condition through sensors, and the Bike Unit, which controls the ignition relay and verifies riding readiness. Communication between the two units is achieved using RF or Bluetooth modules. In addition, the GSM and GPS modules are integrated for accident reporting and location tracking. The system design also incorporated an AI layer for decision-making, ensuring that accident detection is not purely threshold-based but enhanced by machine learning analysis of multi-sensor data.

Step 3: Hardware Development

In this stage, all the required sensors and modules were procured and assembled. The Helmet Unit integrates helmet detection sensors, an MQ-3 gas sensor for alcohol levels, an accelerometer, a gyroscope, and vibration/force sensors for accident detection. The Bike Unit is equipped with a load sensor and an ignition relay to enforce safety before the engine starts. The Arduino/ESP32 microcontroller was programmed to serve as the processing hub, managing sensor data, ignition logic, and communication signals. A rechargeable battery with optional solar charging was used to power the system, ensuring reliability and sustainability for long-term usage.

Step 4: Software & AI Integration

The software development involved programming the microcontroller using Arduino IDE to read sensor values, apply logical conditions, and execute ignition control. GSM modules were configured with AT commands to send SMS alerts, while GPS modules were coded to capture and format location data. For the AI component, accident datasets were created by simulating falls, sudden impacts, and normal riding conditions. Random Forest algorithms were trained to provide probability-based accident detection, while Support Vector Machines (SVM) were used for binary classification of accident vs. non-accident events. The trained models were optimized using TinyML for deployment on embedded hardware, enabling real-time decision-making without reliance on external servers.

Step 5: Testing and Validation

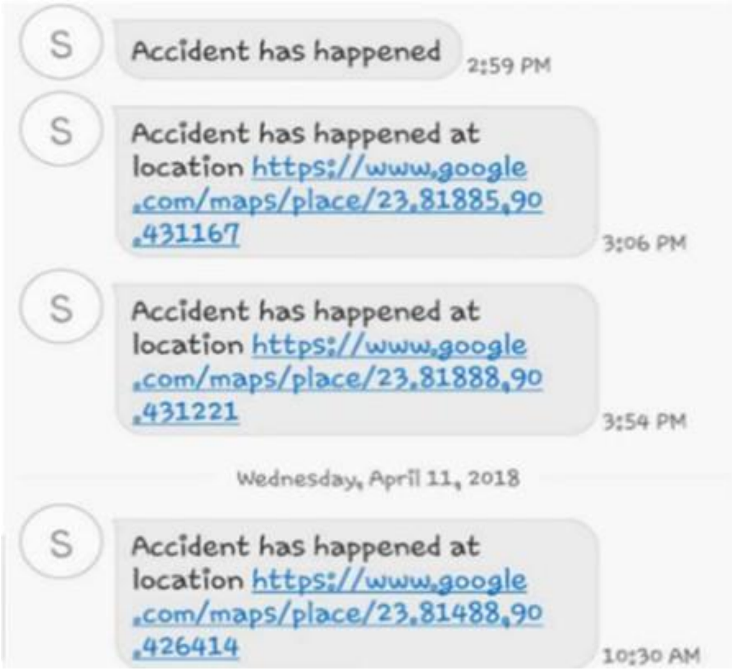
The system was tested in multiple stages. Initially, unit testing was performed to validate each sensor's accuracy—for example, the MQ-3 alcohol sensor was tested against calibrated alcohol levels, and the accelerometer was tested under controlled tilts and shocks. Integration testing ensured smooth communication between the Helmet Unit and Bike Unit for ignition control.

Functional testing was conducted by simulating accidents through controlled impacts to measure system responsiveness. AI models were validated using test datasets, focusing on detection accuracy, false positives, and response time. The GSM and GPS modules were tested for reliability of alert transmission under different network conditions.

Step 6: Deployment & Enhancement

After successful testing, the system was integrated into a standard helmet for field deployment. Road trials were carried out under real riding conditions to confirm reliability and accuracy. During these trials, the system successfully enforced helmet compliance, prevented ignition under alcohol influence, and transmitted emergency alerts in accident scenarios. The final prototype demonstrated the potential of combining IoT and AI for road safety. Future enhancements include linking the helmet to a cloud platform for real-time data storage, predictive analytics for accident prevention, integration with mobile applications for user monitoring, and hardware miniaturization for improved comfort and scalability.

Fig. 9. Helmet Detection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Chapter 6

Conclusion and Future Scope

The AI–IoT based Smart Helmet developed in this project addresses some of the most critical issues in road safety, such as helmet negligence, drunk driving, and delayed emergency response. By integrating sensors with Arduino/ESP32, GSM, and GPS modules, the system ensures that the motorcycle ignition is enabled only when the rider wears a helmet and passes the alcohol test. In addition, real-time monitoring of motion and impact enables the detection of accidents, and emergency alerts with GPS location are automatically transmitted to contacts and authorities. The integration of AI algorithms such as Random Forest and Support Vector Machines further enhances the accuracy of accident classification, significantly reducing false alarms and improving reliability. Overall, the prototype demonstrates a cost-effective and proactive solution for enhancing rider safety, capable of saving lives through timely intervention.

The successful implementation of this system highlights the practicality of combining IoT hardware with AI algorithms for real-world safety applications. The project not only showcases a proof-of-concept but also opens new avenues for smart transportation technologies. With further development and large-scale deployment, this solution has the potential to contribute significantly to reducing accident-related fatalities and improving emergency response systems.

Although the prototype demonstrates strong potential, there are several areas for future improvement and expansion. The system can be enhanced by integrating a cloud-based platform to store and analyze accident data, allowing for predictive analytics and large-scale monitoring. A dedicated mobile application can be developed to provide real-time rider updates, driving behavior analysis, and parental/guardian monitoring features. Incorporating advanced AI models and deep learning algorithms could improve predictive accident detection and enable early warnings in high-risk situations such as overspeeding or drowsiness. Hardware miniaturization and ergonomic design will improve rider comfort and encourage widespread adoption.

Additionally, integrating the system with government traffic databases and emergency services networks could create a more connected and automated road safety ecosystem. With these enhancements, the Smart Helmet can evolve into a comprehensive intelligent transportation safety device capable of significantly reducing road fatalities.

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